infoblox

NETWORK OBSERVABILITY POWERED BY KENTIK

# Kentik Deployment Guide

## Prepared for Contoso Manufacturing

Hybrid network observability rollout

| | |
|---|---|
| **Customer** | Contoso Manufacturing |
| **Project** | Hybrid network observability rollout |
| **Prepared by** | Infoblox Solutions Engineering |
| **Date** | 2026-10-29 |
| **Kentik region** | United States (Ashburn, VA) |
| **Scope** | 2 data center(s), 1 branch(es), AWS, Azure, Synthetic monitoring |

This guide describes the Kentik deployment designed for this environment: what gets installed, which permissions and firewall rules are needed, and the steps to go live. Facts are taken from the Kentik Knowledge Base (kb.kentik.com) and each section names its sources.

Infoblox Confidential. Prepared for the named customer only.

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# 1. Deployment at a glance

Kentik is SaaS. The customer runs a small footprint where the network lives, and everything lands in the customer's Kentik account in the United States (Ashburn, VA) region over encrypted, outbound connections.

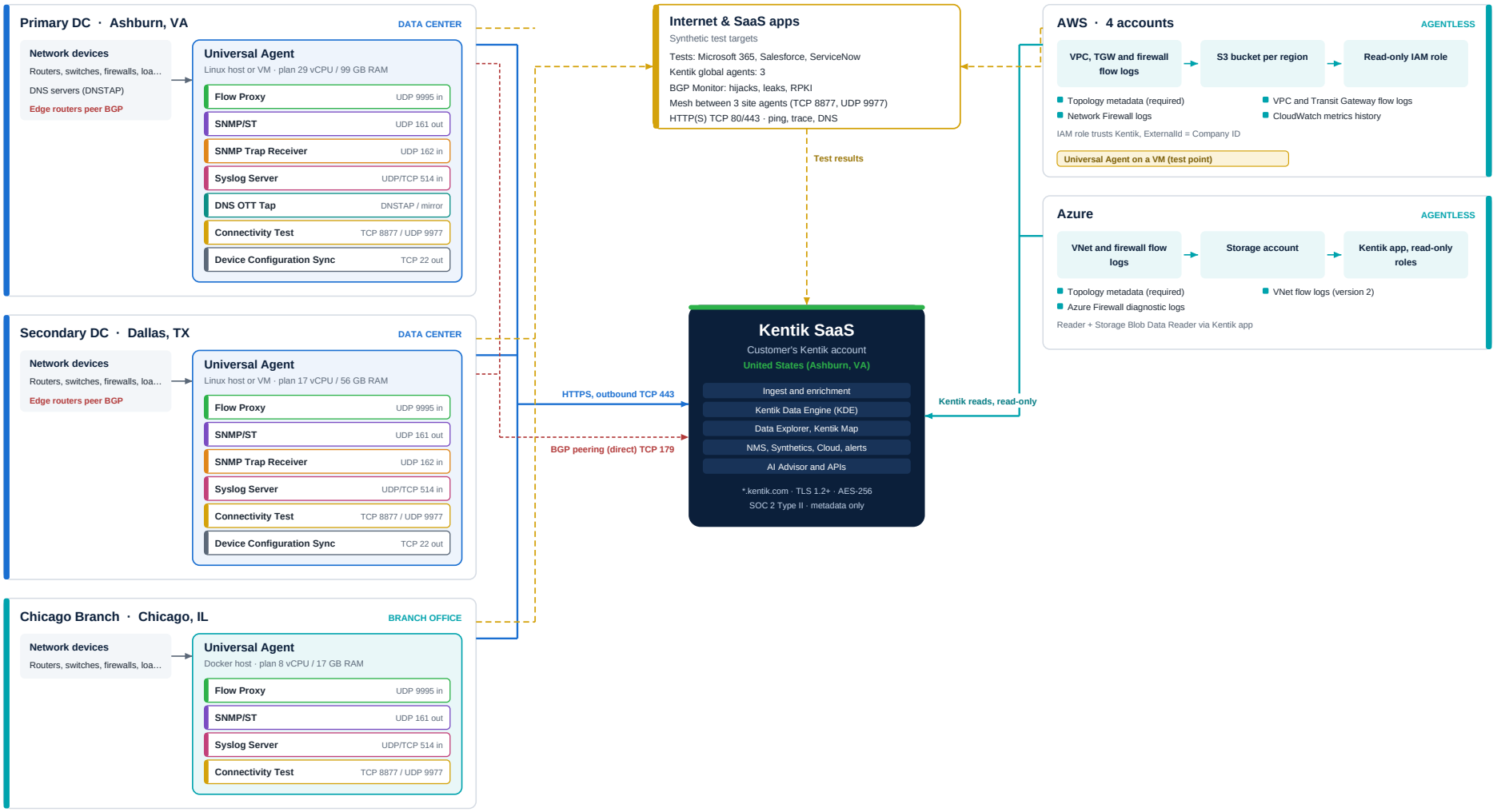
| Where | Deployment | What Kentik collects | Footprint |
|---|---|---|---|
| **Primary DC (Ashburn, VA)** | Universal Agent on Linux | Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync; edge routers peer BGP directly with Kentik | 29 vCPU / 99 GB |
| **Secondary DC (Dallas, TX)** | Universal Agent on Linux | Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync; edge routers peer BGP directly with Kentik | 17 vCPU / 56 GB |
| **Chicago Branch (Chicago, IL)** | Universal Agent on Docker | Flow Proxy Forwarding, SNMP/ST, Syslog Server, Connectivity Test | 8 vCPU / 17 GB |
| **Amazon Web Services** | Agentless (read-only access) | Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs, CloudWatch metrics history | Test agent VM |
| **Microsoft Azure** | Agentless (read-only access) | Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs | No compute |

### Three patterns cover this design

- **On-prem: a Universal Agent per data center.** A single agent with only the capabilities each site needs toggled on in the portal. Devices point at it, and it talks to Kentik over outbound HTTPS. The standalone kproxy agent is not used; the Universal Agent Flow Proxy replaces it.
- **Public cloud: agentless.** Flow logs go to storage in the customer's own account, and Kentik gets read-only access to the logs, topology metadata and metrics.
- **Internet and SaaS: synthetic tests.** Kentik's global agents and the customer's own agents test SaaS apps, websites, DNS and paths between sites and clouds.

Source: Kentik KB, Universal Agents; Public Clouds; Synthetics Overview

## 2. Topology


Primary DC · Ashburn, VA
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
DNS servers (DNSTAP)
Edge routers peer BGP
Universal Agent
Linux host or VM · plan 29 vCPU / 99 GB RAM
Flow Proxy UDP 9995 in
SNMP/ST UDP 161 out
SNMP Trap Receiver UDP 162 in
Syslog Server UDP/TCP 514 in
DNS OTT Tap DNSTAP / mirror
Connectivity Test TCP 8877 / UDP 9977
Device Configuration Sync TCP 22 out
Secondary DC · Dallas, TX
DATA CENTER
Network devices
Routers, switches, firewalls, loa…
Edge routers peer BGP
Universal Agent
Linux host or VM · plan 17 vCPU / 56 GB RAM
Flow Proxy UDP 9995 in
SNMP/ST UDP 161 out
SNMP Trap Receiver UDP 162 in
Syslog Server UDP/TCP 514 in
Connectivity Test TCP 8877 / UDP 9977
Device Configuration Sync TCP 22 out
Chicago Branch · Chicago, IL
BRANCH OFFICE
Network devices
Routers, switches, firewalls, loa…
Universal Agent
Docker host · plan 8 vCPU / 17 GB RAM
Flow Proxy UDP 9995 in
SNMP/ST UDP 161 out
Syslog Server UDP/TCP 514 in
Connectivity Test TCP 8877 / UDP 9977
Internet & SaaS apps
Synthetic test targets
Tests: Microsoft 365, Salesforce, ServiceNow
Kentik global agents: 3
BGP Monitor: hijacks, leaks, RPKI
Mesh between 3 site agents (TCP 8877, UDP 9977)
HTTP(S) TCP 80/443 · ping, trace, DNS
Test results
HTTPS, outbound TCP 443
BGP peering (direct) TCP 179
Kentik SaaS
Customer's Kentik account
United States (Ashburn, VA)
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, Cloud, alerts
AI Advisor and APIs
*.kentik.com · TLS 1.2+ · AES-256
SOC 2 Type II · metadata only
Kentik reads, read-only
AWS · 4 accounts
AGENTLESS
VPC, TGW and firewall flow logs
S3 bucket per region
Read-only IAM role
Topology metadata (required)
Network Firewall logs
VPC and Transit Gateway flow logs
CloudWatch metrics history
IAM role trusts Kentik, ExternalId = Company ID
Universal Agent on a VM (test point)
Azure
AGENTLESS
VNet and firewall flow logs
Storage account
Kentik app, read-only roles
Topology metadata (required)
Azure Firewall diagnostic logs
VNet flow logs (version 2)
Reader + Storage Blob Data Reader via Kentik app
To Kentik over HTTPS
BGP peering
Kentik reads cloud logs and metadata
Synthetic tests
Only outbound connections from customer networks. No inbound access from the Internet.

# 3. Roles and permissions

Who on the customer side does what, and the access each task needs. Kentik only ever receives read-only access.

| Role | Task | Access needed |
|---|---|---|
| **Kentik administrator** | Deploy and authorize Universal Agents, enable capabilities, add devices, create cloud exports and tests | Kentik admin (or super admin) user in the customer's Kentik account |
| **Server or platform team** | Provide 3 agent host(s) and run the install command | root or sudo on a supported Linux host, or Docker |
| **Network team** | Point flow, traps and syslog at the agents; create SNMP and SSH credentials; configure BGP peering | Device configuration rights; SNMP v2c/v3 read-only credentials; read-only SSH account |
| **Security / firewall team** | Open the rules in section 4 | Firewall and proxy change rights |
| **DNS team** | Stream DNSTAP from DNS servers (for example Infoblox NIOS) to the agent | DNS appliance administration |
| **AWS IAM administrator** | AWS: enable flow logs and grant Kentik access | IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket |
| **Entra ID Application Administrator and subscription Owner** | Azure: enable flow logs and grant Kentik access | Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account |

## Access granted to Kentik

| Scope | Access |
|---|---|
| **AWS** | IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket |
| **Azure** | Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account |
| **Network devices** | SNMP read-only (polled from the customer's own agent, never from the Internet) |
| **Network devices** | Read-only SSH from the agent; command block-list, never configure or commit |

Source: Kentik KB, UA Deployment; NMS Setup; SSH Command Access for Kentik NMS; Kentik for AWS; Kentik for Azure

# 4. Network and firewall requirements

All Universal Agent connections are outbound. Nothing needs to reach into the customer network from the Internet.

### Outbound allowlist (every agent)

| Destination | Port | Notes |
|---|---|---|
| flow.kentik.com | TCP 443 | TLS 1.2+ |
| grpc.api.kentik.com | TCP 443 | TLS 1.2+ |
| api.kentik.com | TCP 443 | TLS 1.2+ |
| portal.kentik.com | TCP 443 | TLS 1.2+ |
| storage.googleapis.com | TCP 443 | TLS 1.2+ |
| whoami.kentiklabs.com | TCP 443 | Optional |
| Kentik flow ingest | UDP 40010 | Optional Flow Proxy base port |

- SSL-inspecting firewalls and proxies must allow gRPC over port 443.
- UDP 40010 to Kentik is an optional Flow Proxy base port.
- A web proxy works: set https_proxy before running the install (Linux) or pass it with --env (Docker).

### Primary DC · Ashburn, VA

| From | To | Port | Purpose |
|---|---|---|---|
| Universal Agent | Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com) | TCP 443, TLS 1.2+ | All telemetry, control and updates (outbound only) |
| Universal Agent | Kentik SaaS | UDP 40010 (optional) | Flow Proxy base port |
| Network devices | Universal Agent | UDP 9995 | Flow export (NetFlow, sFlow, IPFIX) |
| Universal Agent | Network devices | UDP 161 | SNMP polling (v1, v2c, v3) |
| Universal Agent | Network devices | ICMP | Reachability |
| Universal Agent | Network devices | Device gNMI port | Streaming telemetry dial-in (only if used) |
| Network devices | Universal Agent | UDP 162 | SNMP traps |
| Network devices: routers, switches, firewalls, load balancers | Universal Agent | UDP and TCP 514 | Syslog |
| DNS servers (for example Infoblox NIOS) | Universal Agent | DNSTAP port you choose | DNSTAP stream, or a SPAN or mirror port |
| Other Universal Agents | Universal Agent (both directions) | TCP 8877 and UDP 9977 | Agent-to-agent and mesh tests |
| Universal Agent | Test targets | ICMP, TCP or UDP | Ping and traceroute |
| Universal Agent | Test targets | TCP 80 and 443 | HTTP(S) and API tests |

| From | To | Port | Purpose |
|---|---|---|---|
| Universal Agent | Device management IPs | TCP 22 | SSH config backup and read-only show commands |
| Edge routers | Kentik ingest peering IPs (shown in device settings) | TCP 179 | BGP peering (direct, telemetry only) |

## Secondary DC · Dallas, TX

| From | To | Port | Purpose |
|---|---|---|---|
| Universal Agent | Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com) | TCP 443, TLS 1.2+ | All telemetry, control and updates (outbound only) |
| Universal Agent | Kentik SaaS | UDP 40010 (optional) | Flow Proxy base port |
| Network devices | Universal Agent | UDP 9995 | Flow export (NetFlow, sFlow, IPFIX) |
| Universal Agent | Network devices | UDP 161 | SNMP polling (v1, v2c, v3) |
| Universal Agent | Network devices | ICMP | Reachability |
| Universal Agent | Network devices | Device gNMI port | Streaming telemetry dial-in (only if used) |
| Network devices | Universal Agent | UDP 162 | SNMP traps |
| Network devices: routers, switches, firewalls, load balancers | Universal Agent | UDP and TCP 514 | Syslog |
| Other Universal Agents | Universal Agent (both directions) | TCP 8877 and UDP 9977 | Agent-to-agent and mesh tests |
| Universal Agent | Test targets | ICMP, TCP or UDP | Ping and traceroute |
| Universal Agent | Test targets | TCP 80 and 443 | HTTP(S) and API tests |
| Universal Agent | Device management IPs | TCP 22 | SSH config backup and read-only show commands |
| Edge routers | Kentik ingest peering IPs (shown in device settings) | TCP 179 | BGP peering (direct, telemetry only) |

## Chicago Branch · Chicago, IL

| From | To | Port | Purpose |
|---|---|---|---|
| Universal Agent | Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com) | TCP 443, TLS 1.2+ | All telemetry, control and updates (outbound only) |
| Universal Agent | Kentik SaaS | UDP 40010 (optional) | Flow Proxy base port |
| Network devices | Universal Agent | UDP 9995 | Flow export (NetFlow, sFlow, IPFIX) |
| Universal Agent | Network devices | UDP 161 | SNMP polling (v1, v2c, v3) |
| Universal Agent | Network devices | ICMP | Reachability |
| Universal Agent | Network devices | Device gNMI port | Streaming telemetry dial-in (only if used) |

| From | To | Port | Purpose |
|---|---|---|---|
| Network devices: routers, switches, firewalls, load balancers | Universal Agent | UDP and TCP 514 | Syslog |
| Other Universal Agents | Universal Agent (both directions) | TCP 8877 and UDP 9977 | Agent-to-agent and mesh tests |
| Universal Agent | Test targets | ICMP, TCP or UDP | Ping and traceroute |
| Universal Agent | Test targets | TCP 80 and 443 | HTTP(S) and API tests |

## Cloud test agents

A Universal Agent on a VM in a cloud network uses the same outbound TCP 443 rules. Allow TCP 8877 and UDP 9977 between it and the other agents in security groups or NSGs.

## Azure storage firewall

Only if the log storage account firewall is restricted: allow 20.69.189.228 and 20.69.185.115 (WestUS2: 51.8.214.254 and 172.171.46.16).

Source: Kentik KB, Universal Agents; UA Deployment; Test Types Reference; Azure Prerequisites & Roles

# 5. Universal Agent deployment

Install one Universal Agent per site, then switch capabilities on in the portal. The agent updates itself and reports its own health.

### Host requirements

- Supported OS: Debian 11, 12, 13; Ubuntu 20.04, 22.04, 24.04; Oracle Linux 9; CentOS Stream 8. CentOS 7 and Enterprise Linux before 8 are not supported.
- Or any Docker host that can run the generated docker run command.
- Size for the sum of the capabilities on the host, plus 20 to 30% headroom for bursty flow, syslog and traps. The plans below use 30%.
- Disk sizing is not published in the KB; confirm with Kentik.

> **Rolling out many agents?** Automated rollout (optional): create a provisioning token with POST https://grpc.api.kentik.com/kagent/v202401/provisioning-tokens (name, maxUsageCount, requiresApproval, config.siteId, optional expiresAt), export K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN on the host, then authorize with PUT .../agents/authorize/<install-id>. A token created with requiresApproval false skips authorization. Revoke the token when all agents have registered.

Source: Kentik KB, Universal Agents; UA Deployment

## 5.1 Primary DC · Ashburn, VA

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync**.

| Capability | Planned load | CPU / RAM |
|---|---|---|
| Flow Proxy Edge Processing | 40,000 flows/s (scaled from the KB figure: confirm with Kentik) | 8 cores / 32 GB x 2 |
| SNMP/ST | 350 devices | 2 cores / 8 GB |
| Syslog Server | 3,000 events/s | 1 core / 1 GB |
| DNS OTT Tap | 50,000 lookups/s | 1 core / 1 GB |
| Connectivity Test | per agent | 2 cores / 2 GB |
| SNMP Trap Receiver | event driven | Not published |
| Device Configuration Sync | per fetch interval | Not published |
| Total from KB sizing | | 22 cores / 76 GB |
| Recommended host (+30%) | | 29 vCPU / 99 GB RAM |

### Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.

4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Primary DC), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, DNS OTT Tap, Connectivity Test, Device Configuration Sync) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

### BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

## 5.2 Secondary DC · Dallas, TX

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync**.

| Capability | Planned load | CPU / RAM |
|---|---|---|
| Flow Proxy Edge Processing | 15,000 flows/s | 8 cores / 32 GB |
| SNMP/ST | 180 devices | 2 cores / 8 GB |
| Syslog Server | 1,500 events/s | 1 core / 1 GB |
| Connectivity Test | per agent | 2 cores / 2 GB |
| SNMP Trap Receiver | event driven | Not published |
| Device Configuration Sync | per fetch interval | Not published |
| Total from KB sizing | | 13 cores / 43 GB |
| Recommended host (+30%) | | 17 vCPU / 56 GB RAM |

### Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Secondary DC), then click Continue.

6. Toggle on the capabilities for this site (Flow Proxy Edge Processing, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, Device Configuration Sync) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

Behind a web proxy run export https_proxy=http://<proxy>:<port> before the install command.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
export https_proxy=http://<proxy>:<port>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

### BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

## 5.3 Chicago Branch · Chicago, IL

Branch office. Install method: **Docker**. Capabilities: **Flow Proxy Forwarding, SNMP/ST, Syslog Server, Connectivity Test**.

| Capability | Planned load | CPU / RAM |
|---|---|---|
| Flow Proxy Forwarding | 2,000 flows/s | 1 core / 2 GB |
| SNMP/ST | 25 devices | 2 cores / 8 GB |
| Syslog Server | 200 events/s | 1 core / 1 GB |
| Connectivity Test | per agent | 2 cores / 2 GB |
| Total from KB sizing | | 6 cores / 13 GB |
| Recommended host (+30%) | | 8 vCPU / 17 GB RAM |

### Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Docker. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the container ID). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Chicago Branch), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy Forwarding, SNMP/ST, Syslog Server, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
# run the docker command copied from Settings > Universal Agents > Deploy Agent
# it carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN
```

Source: Kentik KB, UA Deployment; UA Portal Management; Device Settings

# 6. Capability configuration

How to configure each capability used in this design, on the agent and on the devices that feed it.

## Flow Proxy

Receives NetFlow, sFlow and IPFIX from routers, switches and firewalls and sends it to Kentik over TLS. Forwarding sends the raw flow to Kentik SaaS, which decodes it, so it needs far less CPU and RAM. Edge Processing decodes and enriches the flow on the agent before sending it, so it needs a much larger host.

**Enabled at:** Primary DC (Edge Processing), Secondary DC (Edge Processing), Chicago Branch (Forwarding) · **License:** FlowPak

- Toggle on Flow Proxy on the agent in the mode chosen for this site (Forwarding or Edge Processing). Settings: Host 0.0.0.0, Port 9995 (UDP), optional sFlow agent address override, and Site.
- On each device, export flow to the agent's IP on UDP 9995. Sample ingress on every interface; never ingress and egress on the same interface.
- The sample rate on the device must match the rate set for that device in Kentik.
- Vendor configuration snippets for flow, SNMP and BGP: github.com/kentik/config-snippets.
- Sanity check after go-live: flow volume should land within about 20% of SNMP interface traffic.

Source: Kentik KB, Universal Agents; Router Configuration; Flow Overview; Migration Scenarios (kproxy to UA)

## SNMP/ST

Discovers devices and collects health, interface and hardware metrics by SNMP polling or gNMI streaming telemetry. It also enriches flow with interface names and speeds.

**Enabled at:** Primary DC, Secondary DC, Chicago Branch · **License:** FlowPak (flow enrichment) or NMS / DevicePak (full monitoring)

- Run the NMS Setup Wizard: pick this agent, enter IP ranges and SNMP v1, v2c or v3 credentials, discover, then add devices.
- Per device, on the SNMP tab choose "Flow Enrichment via Universal Agent" or "Full Monitoring via Universal Agent".
- SNMP v3 options: auth None, MD5 or SHA; privacy None, DES or AES.
- Default polling: 1 minute for status, CPU, memory and interfaces; 5 minutes for everything else.
- Streaming telemetry (optional): enable "Dial-In from Universal Agent" (gNMI over gRPC, 30 second default).

Source: Kentik KB, Universal Agents; NMS Setup; Device Settings

## SNMP Trap Receiver

Captures SNMP traps the moment a device sends them, for example an interface or BGP session going down.

**Enabled at:** Primary DC, Secondary DC · **License:** NMS / DevicePak

- Enable SNMP Trap Receiver on the agent.
- Point device trap targets at the agent's IP on UDP 162.
- Traps must come from a device licensed for NMS. Query them in Data Explorer with the metric "SNMP Traps: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

## Syslog Server

Collects log messages from network devices (routers, switches, firewalls, load balancers) so they can be searched, alerted on and lined up with traffic.

**Enabled at:** Primary DC, Secondary DC, Chicago Branch · **License:** NMS / DevicePak

- Enable Syslog Server on the agent.
- Point network device syslog at the agent's IP on UDP or TCP 514.
- If a syslog forwarder sits in between, it must keep the original device IP. Changing the port currently needs Kentik support.
- Query in Data Explorer with the metric "Syslog: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

## DNS OTT Tap

Turns DNS lookups into records Kentik can join with traffic, for OTT service tracking and DNS troubleshooting.

**Enabled at:** Primary DC · **License:** Confirm with the Kentik account team

- Enable DNS OTT Tap on the agent.
- Preferred mode: DNSTAP receiver, for DNS appliances such as Infoblox. Other modes: promiscuous capture of a SPAN or mirror port, or directly on a BIND9 or Linux resolver.
- On the DNS servers, stream DNSTAP to the agent's IP and the port you configure.

Source: Kentik KB, Universal Agents

## Connectivity Test

Makes this agent a private test point. It measures latency, loss and jitter to other sites, clouds, SaaS apps and DNS servers, and gives AI Advisor on-demand ping and traceroute.

**Enabled at:** Primary DC, Secondary DC, Chicago Branch · **License:** Synthetics test credits

- Toggle on Connectivity Test on the agent.
- Open TCP 8877 and UDP 9977 between every pair of agents that will test each other.
- Create tests in Synthetics: agent-to-agent, network mesh, server IP or hostname, DNS server monitor and HTTP(S) or API tests.
- Confirm which continuous test types run on the Universal Agent today with the Kentik account team.

Source: Kentik KB, Universal Agents; Synthetics Overview; Test Types Reference; Test Credits

## Device Configuration Sync

Backs up configs, shows what changed, and runs read-only show commands during an AI Advisor investigation.

**Enabled at:** Primary DC, Secondary DC · **License:** NMS / DevicePak

- Create a read-only SSH account on each device (TACACS+ or RADIUS logs will show the agent as the source).
- In device settings, SSH tab: credential, collection agent, platform autodetect, config sync and fetch interval (5 minutes recommended for high-visibility environments).
- Supported platforms: Juniper Junos, Cisco NX-OS, Arista EOS and Cisco IOS-XE.
- Read-only by design: a command block-list, never configure or commit.

Source: Kentik KB, SSH Command Access for Kentik NMS; Device Settings

# 7. Public cloud setup

No agents are needed in public clouds. Each cloud writes flow logs to storage in the customer's own account, and Kentik reads them with read-only access. Kentik recommends a metadata-only export first to prove access before turning on high-volume logs.

## Amazon Web Services

No agent. Flow logs are written to S3 buckets the customer owns. Kentik's collector, running in Kentik's own AWS account, reads them inside AWS and sends them on to Kentik SaaS, so the customer pays no egress for the logs. Metadata API calls come from Kentik SaaS through a read-only IAM role, protected by an ExternalId.

**Collecting:** Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs, CloudWatch metrics history. Scope: 4 account(s)/subscription(s)/project(s). Regions: us-east-1, us-west-2.

### Steps

1. Start with a metadata-only export to prove access before turning on high-volume logs.
2. Create one S3 bucket per region for flow logs. Kentik treats each bucket as one device, so hundreds of VPCs into one global bucket risks S3 throttling.
3. Encryption: SSE-S3 is recommended. SSE-KMS needs the KMS statement. SSE-C is not supported.
4. Turn on flow logs for VPCs, Transit Gateways and Network Firewalls: filter All, destination S3, custom format with every v2 to v5 field.
5. Create the IAM role with the trust policy and permissions below.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > AWS Cloud. Select features, enter the Role ARN > Verify Role, Region > Verify Region, bucket name and prefix > Verify S3 Bucket.
7. Pick sampling (Legacy, a rate from 2 to 2000, or Unsampled), name the export, choose the billing plan and Save.
8. Optional: tick "Help me configure my provider via Terraform" for a main.tf that creates the role, bucket and flow logs.

### How data moves

- Flow logs: Kentik's collector runs in Kentik's AWS account (834693425129). It assumes your role and reads new log files from your S3 buckets inside AWS, then sends them to Kentik SaaS, so you don't pay egress charges for the logs.
- Metadata: Kentik SaaS calls the AWS describe APIs (EC2, Direct Connect, ELB, Network Firewall and so on) through the same read-only role, using sts:AssumeRole with your ExternalId.

### IAM role trust policy (primary account)

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::834693425129:root" },
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": { "sts:ExternalId": "<Kentik Company ID>" },
      "ArnEquals": { "aws:PrincipalArn": "arn:aws:iam::834693425129:role/eks-ingest-node" }
    }
  }]
}
```

### Nested accounts: primary and secondary roles

Kentik's account assumes the primary role in account A. The primary role then assumes a secondary role in each other account (B, C and so on) to read its metadata.

1. Pick the primary (hub) account, for example the account that holds the log buckets. Kentik assumes only the primary role.
2. In the primary account, create the primary role with the trust policy above (Kentik's account, ExternalId = Company ID).
3. Give the primary role a policy with sts:AssumeRole on the secondary roles ("AssumeSecondaryRoles") and organizations:ListAccounts.
4. In each secondary account, create a secondary role that trusts the primary account (arn:aws:iam::<primary account ID>:root, sts:AssumeRole, no ExternalId) and attach the read-only metadata permissions below. Use the same role name in every account.
5. In the Kentik export, tick "This is an AWS organization role", then enter the secondary account IDs, their regions and the secondary role name.
6. Flow logs from the secondary accounts can be sent to a bucket in the primary account (a central log archive), so one export reads them all.

### Secondary role trust policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::<primary account ID>:root" },
    "Action": "sts:AssumeRole",
    "Condition": {}
  }]
}
```

### IAM permissions policy

| Service | Actions |
|---|---|
| **EC2** | ec2:Describe*, ec2:Search*, ec2:GetManagedPrefixListEntries |
| **Direct Connect** | directconnect:List*, directconnect:Describe* |
| **ELB** | elasticloadbalancing:Describe* |
| **IAM** | iam:ListAccountAliases |
| **Network Firewall** | network-firewall:Describe*, network-firewall:List* |
| **Cloud WAN** | networkmanager:ListCoreNetworks, GetCoreNetwork, GetCoreNetworkPolicy, ListAttachments, GetNetworkRoutes |
| **CloudWatch** | cloudwatch:ListMetrics, GetMetricStatistics, GetMetricData, Describe* |
| **Organizations** | organizations:ListAccounts |
| **S3 (log bucket)** | s3:Get*, s3:List* on arn:aws:s3:::<bucket> and arn:aws:s3:::<bucket>/* |
| **KMS (SSE-KMS only)** | kms:Decrypt, kms:DescribeKey on the bucket key |

Export status values: Start, Pending, Error, OK, Disabled or Halt. Licensing: CloudPak, measured in flows per second.

**Test agent in AWS:** install a Universal Agent on a small VM with the Synthetics capability, using the same steps as on-prem. It needs outbound TCP 443 only, plus TCP 8877 and UDP 9977 to the other agents.

Source: Kentik KB, Kentik for AWS; Metadata Configuration (AWS); Flow/Firewall Log Collection Configuration (AWS); Kentik Export Configuration for AWS; Public Clouds

## Microsoft Azure

No agent. Network Watcher writes VNet flow logs to a storage account the customer owns, and Kentik reads it through an app approved in Entra ID.

**Collecting:** Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs.

### Steps

1. Create a dedicated storage account for the logs.
2. If the storage account firewall is restricted, allow 20.69.189.228 and 20.69.185.115 (WestUS2 uses 51.8.214.254 and 172.171.46.16).
3. In Network Watcher turn on VNet flow logs (version 2) to that storage account.
4. Send Azure Firewall diagnostic logs to the same storage account with the azfwapplicationrule and azfwnetworkrule categories.
5. Authorize Kentik: an Application Administrator assigns the Kentik enterprise app (Kentik NSG Flow Exporter), then Validate Service Principal in the export wizard.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > Azure Cloud. Pick features, API access method, location and subscription ID, Validate Service Principal, resource group > Verify Access.
7. Enter the storage account name, sampling (2 to 2000 or Unsampled), optional enrichment scope and billing plan, then Save.
8. Optional: the export wizard can generate a PowerShell script that does most of the Azure side (VNet flow logs only).

### Roles for the Kentik service principal

| Role | Scope | Why |
|---|---|---|
| **Reader** | Subscription or resource group holding the logs | Read resources |
| **Reader** | Additional subscriptions or resource groups | Enrichment scope |
| **Storage Blob Data Reader (or Reader and Data Access)** | Log storage account | Read exported logs |
| **Custom read-only Kentik role (optional)** | Tenant | Full traffic path views |

Source: Kentik KB, Kentik for Azure; Azure Prerequisites & Roles; Flow/Firewall Log Collection (Azure); Kentik Export Configuration for Azure; Automated Configuration (PowerShell)

# 8. Synthetic monitoring plan

Continuous tests measure latency, loss and jitter between sites, clouds and the applications users depend on.

## Test points

| Agent | Type |
|---|---|
| Primary DC | Private (Universal Agent) |
| Secondary DC | Private (Universal Agent) |
| Chicago Branch | Private (Universal Agent) |
| AWS test agent | Private (Universal Agent) |
| 3 Kentik global agent(s) | Kentik-hosted: Internet hubs, cloud regions, app agents |

## Planned tests

- Network mesh between all private agents (latency, loss, jitter between every pair of sites and clouds).
- HTTP(S) test to Microsoft 365 from private and global agents.
- HTTP(S) test to Salesforce from private and global agents.
- HTTP(S) test to ServiceNow from private and global agents.
- BGP Monitor for the customer's prefixes: hijacks, leaks and RPKI status.
- Available test types: Agent-to-Agent, Network Mesh, Server IP, Server Hostname, Network Grid, autonomous tests (ASN, CDN, country, region, city), DNS Server Monitor, DNS Server Grid, HTTP(S) or API, Page Load and Transaction.

## Estimated monthly test credits

| Test | Interval | Per run | Credits / month |
|---|---|---|---|
| Mesh between 4 private agents | every 1 min | 4 x 3 subtests x 2.5 = 30 per run | 1,296,000 |
| HTTP(S) tests to 3 SaaS app(s) | every 5 min | 4 private + 3 global agents, 5 s timeout = 75 per run | 648,000 |
| BGP Monitor (1 test) | continuous | 0.53 credits per minute | 22,896 |
| Total | | | 1,966,896 |

Credit rules from the KB: network subtest 2.5 credits from a private agent and 5 from a global agent; HTTP subtests cost timeout (seconds) x 0.5 private or x 1 global; BGP Monitor 0.53 credits per minute; a test every minute runs 43,200 times in a 30-day month; Test Preview is free until saved. Mesh estimates count each direction as one subtest (the KB example: a 4-agent private mesh is 30 credits per run). Treat these as planning estimates and confirm with Kentik.

Source: Kentik KB, Synthetics Overview; Test Types Reference; Test Credits; Agents Page

# 9. Go-live validation checklist

| | |
|---|---|
| ☐ | Primary DC: agent shows Up in Settings > Universal Agents, capabilities show enabled |
| ☐ | Primary DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic |
| ☐ | Primary DC: devices discovered in NMS with interfaces and health metrics |
| ☐ | Primary DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events) |
| ☐ | Primary DC: first config backup fetched over SSH |
| ☐ | Primary DC: BGP sessions established with Kentik; flows show AS path |
| ☐ | Secondary DC: agent shows Up in Settings > Universal Agents, capabilities show enabled |
| ☐ | Secondary DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic |
| ☐ | Secondary DC: devices discovered in NMS with interfaces and health metrics |
| ☐ | Secondary DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events) |
| ☐ | Secondary DC: first config backup fetched over SSH |
| ☐ | Secondary DC: BGP sessions established with Kentik; flows show AS path |
| ☐ | Chicago Branch: agent shows Up in Settings > Universal Agents, capabilities show enabled |
| ☐ | Chicago Branch: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic |
| ☐ | Chicago Branch: devices discovered in NMS with interfaces and health metrics |
| ☐ | Chicago Branch: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events) |
| ☐ | AWS: cloud export status OK; metadata visible in Kentik Map; flow logs arriving |
| ☐ | Azure: cloud export status OK; metadata visible in Kentik Map; flow logs arriving |
| ☐ | Synthetics: tests show Healthy, Warning or Critical (not Pending); agents Active |
| ☐ | Alerting: at least one alert policy routed to the customer's notification channel |

Source: Kentik KB, UA Portal Management; Flow Overview; SNMP Traps & Syslog; Public Clouds

# 10. Security, data handling and licensing

| Area | Detail |
|---|---|
| **Hosting** | US: Ashburn, VA. EU: Frankfurt. Chosen when the contract is signed. |
| **In transit** | TLS 1.2+ to and from Kentik |
| **At rest** | AES-256, consistent with FIPS 140-2 |
| **Isolation** | Separate logical database and tables per customer |
| **Audits** | SOC 2 Type II yearly, annual third-party penetration test |
| **Access** | SAML 2.0 SSO, MFA (TOTP or hardware keys), user, admin and super admin roles, per-user API tokens |
| **Data** | IP addresses and network metadata, never packet payloads; IP redaction available |
| **Retention (default)** | 45 days full-resolution flow, 120 days summarized, 3 or 13 months NMS metrics by plan; contracts can change this |

Source: Kentik KB, Security Overview; KDE Overview

## Licensing

| License | Covers |
|---|---|
| **FlowPak** | Flow from routers, switches, firewalls and hosts, including SNMP flow enrichment. |
| **NMS / DevicePak** | Full device monitoring, plus syslog, SNMP traps and SSH config and commands. |
| **CloudPak** | Public cloud flow logs, sized in flows per second. |
| **Synthetics** | Monthly test credits: global agent subtest 5 credits, private agent 2.5, BGP Monitor 0.53 per minute. |

*Plans and editions vary by contract. Confirm entitlements with the Kentik account team.*

Source: Kentik KB, Licenses; Test Credits

# 11. Open items to confirm with Kentik

- Which continuous synthetic test types run on the Universal Agent today (the KB names the capability Connectivity Test).
- Licensing names and entitlements per contract (FlowPak, NMS/DevicePak, CloudPak, Synthetics credits).
- Disk sizing for Universal Agent hosts is not published in the KB.
- BGP Proxy shows in Installed Capabilities in the portal, but the KB page still lists it as coming soon. Confirm it is available for this account before relying on it.
- Flow Proxy Forwarding (ktraffic) is marked beta in the KB. Confirm the mode (Forwarding or Edge Processing) per site. Above the KB flow figures, sizing is scaled linearly and is an estimate.

# 12. Appendix: Kentik KB references

| Page | URL |
|---|---|
| Universal Agents | https://kb.kentik.com/docs/universal-agents |
| Public Clouds | https://kb.kentik.com/docs/public-clouds |
| Synthetics Overview | https://kb.kentik.com/docs/synthetics-overview |
| UA Deployment | https://kb.kentik.com/docs/ua-deployment |
| NMS Setup | https://kb.kentik.com/docs/nms-setup |
| SSH Command Access for Kentik NMS | https://kb.kentik.com/docs/ssh-command-access-for-kentik-nms |
| Kentik for AWS | https://kb.kentik.com/docs/kentik-for-aws |
| Kentik for Azure | https://kb.kentik.com/docs/kentik-for-azure |
| Test Types Reference | https://kb.kentik.com/docs/test-types |
| Azure Prerequisites & Roles | https://kb.kentik.com/docs/azure-prerequisites-roles |
| UA Portal Management | https://kb.kentik.com/docs/ua-portal-management |
| Device Settings | https://kb.kentik.com/docs/device-settings |
| Router Configuration | https://kb.kentik.com/docs/router-configuration |
| Flow Overview | https://kb.kentik.com/docs/flow-overview |
| Migration Scenarios (kproxy to UA) | https://kb.kentik.com/docs/kproxy-migration-scenarios |
| SNMP Traps & Syslog | https://kb.kentik.com/docs/snmp-traps-syslogs |
| Test Credits | https://kb.kentik.com/docs/test-credits |
| Metadata Configuration (AWS) | https://kb.kentik.com/docs/metadata-configuration-aws |
| Flow/Firewall Log Collection Configuration (AWS) | https://kb.kentik.com/docs/flow-firewall-log-collection-configuration |
| Kentik Export Configuration for AWS | https://kb.kentik.com/docs/kentik-export-configuration-aws |
| Flow/Firewall Log Collection (Azure) | https://kb.kentik.com/docs/flow-firewall-log-collection |
| Kentik Export Configuration for Azure | https://kb.kentik.com/docs/kentik-export-configuration-azure |
| Automated Configuration (PowerShell) | https://kb.kentik.com/docs/automated-configuration-powershell |
| Agents Page | https://kb.kentik.com/docs/agents-page |
| Security Overview | https://kb.kentik.com/docs/security-overview |
| KDE Overview | https://kb.kentik.com/docs/kde-overview |
| Licenses | https://kb.kentik.com/docs/licenses |

Facts in this guide were checked against the Kentik Knowledge Base in October 2026. Kentik updates its product frequently; confirm details against the linked pages before go-live.